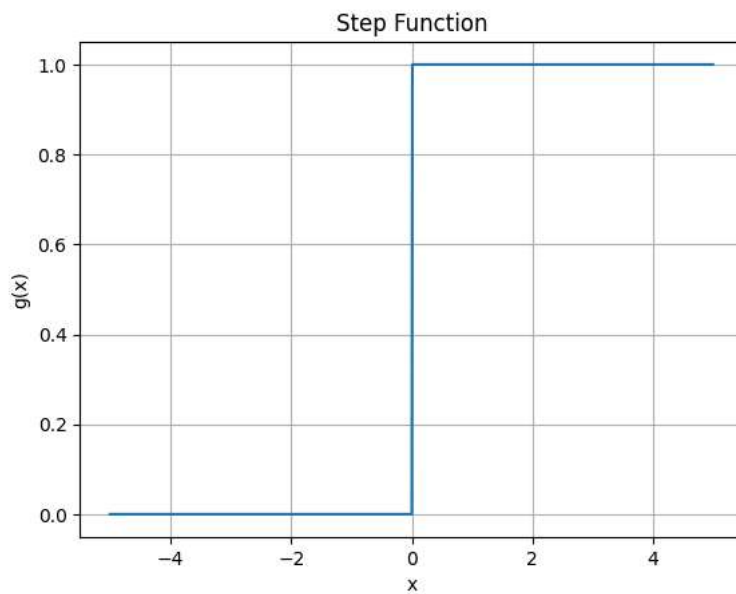


```
import numpy as np
import matplotlib.pyplot as plt

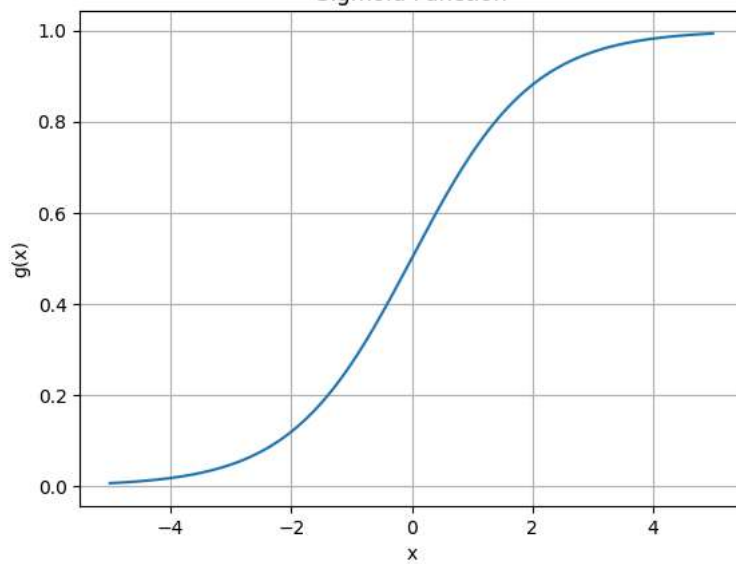
x = np.arange(-5, 5, 0.01)
y = np.where(x < 0, 0, 1)

plt.plot(x, y)
plt.title('Step Function')
plt.xlabel('x')
plt.ylabel('g(x)')
plt.grid()
plt.show()
```



```
def sigmoid(x):
    return 1 / (1 + np.exp(-x))
x = np.arange(-5, 5, 0.01)
plt.plot(x, sigmoid(x))
plt.title('Sigmoid Function')
plt.xlabel('x')
plt.ylabel('g(x)')
plt.grid()
plt.show()
```

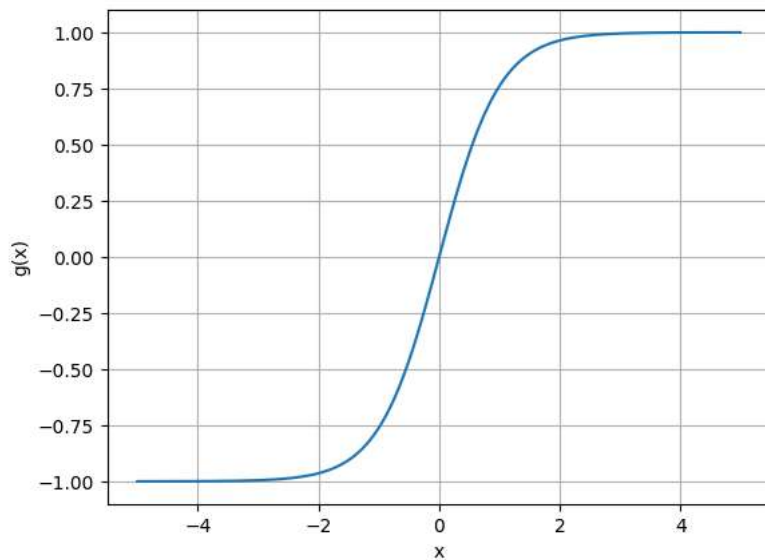
Sigmoid Function



```
def tanh(x):
    return (np.exp(x) - np.exp(-x)) / (np.exp(x) + np.exp(-x))

x = np.arange(-5, 5, 0.01)
plt.plot(x, tanh(x))
plt.title('Tanh')
plt.xlabel('x')
plt.ylabel('g(x)');
plt.grid()
plt.show()
```

Tanh



```
x = np.arange(-5, 5, 0.01)
y = np.maximum(0, np.where(x < 0, 0, x))

plt.plot(x, y)
plt.title('ReLU')
plt.xlabel('x')
plt.ylabel('g(x)')
plt.grid()
plt.show()
```

ReLU

